Cambridge (CIE) A Level Computer Science

Your notes

Algorithms

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- Linear Search
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Linear search

What is a linear search?

- The linear search is a standard algorithm used to find elements in an **unordered list**
- The list is searched **sequentially** and **systematically** from the start to the end one element at a time
- **Comparing each element** to the **value being searched** for
  - If the **value is found** the algorithm **outputs where it was found in the list**
  - If the **value is not found** it **outputs a message stating it is not in the list**
- An example of using a linear search would be looking for a specific student name in a list or searching for a supermarket item in a shopping list

Performing the linear search

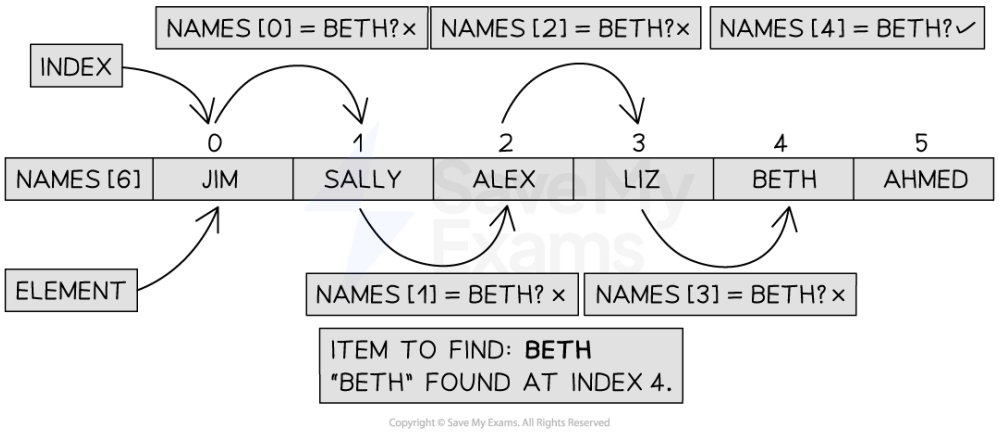

Figure 1: Performing the Linear Search

Time complexity of a linear search

- To determine the algorithm's execution time as measured by the number of instructions or steps carried out, Big-O notation must be used
- The loop is executed n times, once for each item. There are two instructions within the loop, the if statement and an assignment, hence the Big-O for the loop is O(2n)
- There are three initial statements which are O(1)
- The overall Big-O notation is therefore 2n + 3, which when coefficients are factored out, becomes **O(n) as linear time dominates constant time**. The **constant term and coefficients contribute significantly less** as the input size n grows larger

- It is important to remember here that **O(n) is the worst case scenario**. The **best case scenario O(1)** would find the item the first time, while the **average case** would be O(n/2) which when we remove coefficients still becomes **O(n)**

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Space complexity of a linear search

- As the linear search requires no additional space, it is space complexity O(n) where n is the number of items in the list

Tracing a linear search

- Given the following list [5, 4, 7, 1, 3, 8, 9, 2], a trace table for the linear search is shown below

Trace Table of the Linear Search

item	index	i	list[i]	found
8	-1	0	5	False
		1	4	
		2	5	
		3	1	
		4	3	
8	5	5	8	True

Implementing a linear search

Pseudocode

```
FUNCTION linearSearch(list : ARRAY OF STRING, item : STRING) RETURNS INTEGER
   DECLARE index : INTEGER
   DECLARE i : INTEGER
   DECLARE found : BOOLEAN

   index ← -1
   i ← 0
   found ← FALSE

   WHILE i < LENGTH(list) AND found = FALSE
      IF list[i] = item THEN
         index ← i
         found ← TRUE
      ENDIF
      i ← i + 1
   ENDWHILE

   RETURN index
ENDFUNCTION
```

- **Function definition**:
  FUNCTION linearSearch(list : ARRAY OF STRING, item : STRING) RETURNS INTEGER
  - The function takes a list and a target item.
  - It returns the **index** of the item if found, or **–1** if not.
- **Variable declarations**:
  index, i, and found are declared:
  - index stores the result (default is -1 if not found).
  - i is the loop counter.
  - found is a boolean used to stop the search once the item is found.
- **Initialisation**:
  - index ← -1 → start with "not found"
  - i ← 0 → start from the first index of the list
  - found ← FALSE → the item hasn't been found yet
- **Loop condition**:
  WHILE i < LENGTH(list) AND found = FALSE
  - Continue looping **while** there are items left to check **and** the item hasn't been found.
- **Comparison inside loop**:
  IF list[i] = item THEN
  - Check if the current element equals the search item.
- **If a match is found**:
  - index ← i → save the current index
  - found ← TRUE → stop searching
- **Increment**:
  - i ← i + 1 → move to the next item
- **End of loop**:
  - Once the loop finishes (either item is found or end of list is reached), return the result.
- **Return value**:
  RETURN index
  - Returns the index where the item was found.
  - If it wasn't found, it returns -1.

Python

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```
def linear_search(list, item):
    index = -1
    for i in range(len(list)):
        if list[i] == item:
            index = i
            break  # Stop the loop once the item is found
    return index
```

Java

```
public class LinearSearch {
    public static int linearSearch(int[] list, int item) {
        int index = -1;
        for (int i = 0; i < list.length; i++) {
            if (list[i] == item) {
                index = i;
                break;  // Stop the loop once the item is found
            }
        }
        return index;
    }
```

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Binary search

What is a binary search?

- The binary search is a more efficient search method than linear search
- The binary search **compares the middle item to the searched for target item**. If the **values match** then the **index is returned**. If not then the list is adjusted such that:
  - the **top half is ignored** if the **target item is less** than the middle value
  - or the **bottom half is ignored** if the **target item is larger** than the middle value
- Once the list has been adjusted, the search continues until the item is found or the item is not in the list
- It is important to note that the **list must be sorted** for the binary search to work correctly
- The implementation of the binary search is shown below

Performing the binary search

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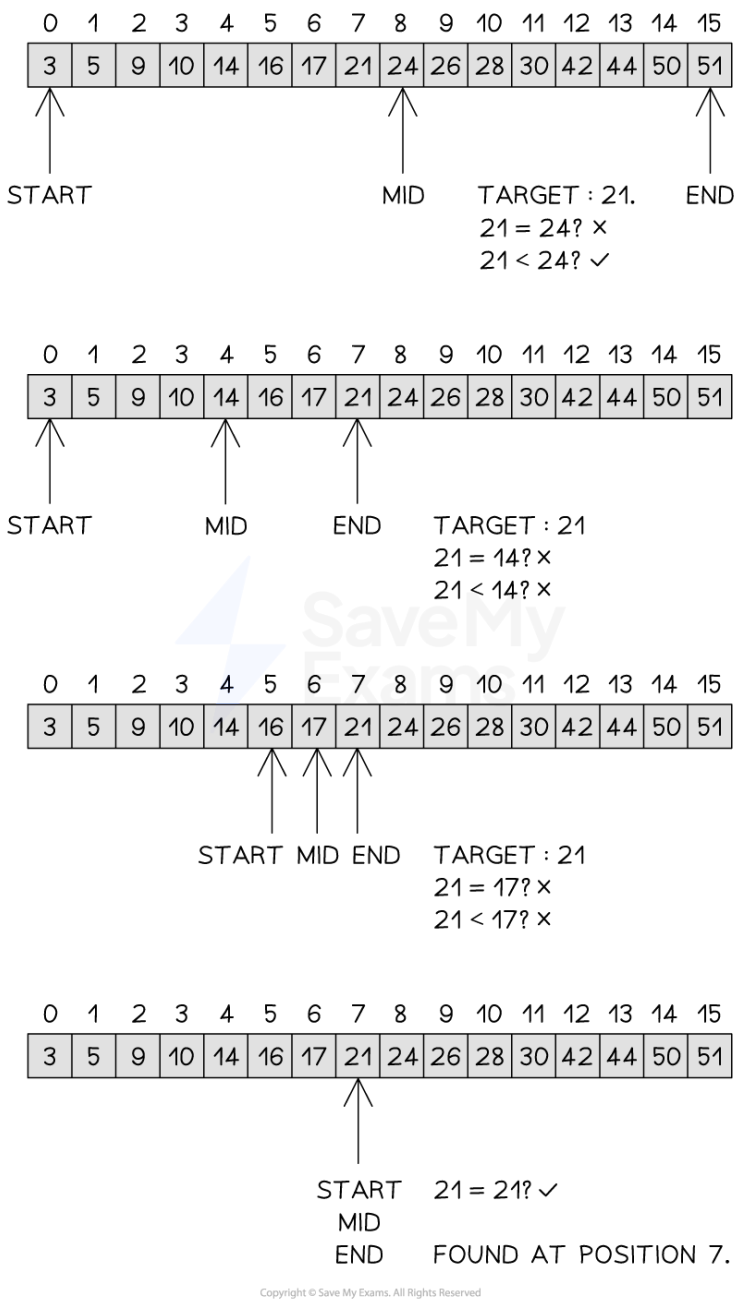

Figure 2: Performing the Binary Search

Examiner Tips and Tricks

- When determining the rounding of the midpoint, rounding up or down are both valid provided consistent use of the same rounding is used throughout the algorithm
- It is important to note that the binary search does not discard, delete or remove parts of the list, it only adjusts the start, end and mid pointers. This only gives the

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appearance that items have been discarded or deleted

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Time complexity of a binary search

- The binary search is an example of a logarithmic time complexity O(logn) as it halves the search space with each iteration of the while loop. This approach is known as **divide and conquer**, where a problem is progressively reduced in size such that when each problem is at its smallest it is easiest to solve
- The algorithm starts with n items before the first iteration. After the first there are n/2 items, after the second there are n/4 items, after the third there are n/8 items and so on, leading to $n/2^i$ after i iterations
- The **worst case scenario** or maximum number of iterations to find one item is where $n/2^i = 1$
  - Multiply both sides by 2^i to get: $n = 2^i$
  - Take the logarithm of both sides: $\log n = \log 2^i$
  - Rewrite with i at the front (using laws of logarithms): $\log n = i \log 2$
  - Log2 (assuming a base of 2 is used) becomes 1, giving: $\log n = i$
  - The binary search is therefore **O(logn)**
- The **best case scenario would be O(1)** where the item is found on the first comparison, while the **average case** would be O(logn/2) which would find the item somewhere in the middle of the search. Removing coefficients means the average case would therefore still be **O(logn)**

Space complexity of a binary search

- As the binary search requires no additional space, it is space complexity O(n), where n is the number of items in the list

Tracing a binary search

- Given the following list [3, 5, 9, 10, 14, 16, 17, 21, 24, 26, 28, 30, 42, 44, 50, 51], a trace table for the binary search is shown below

Trace Table for the Binary Search

item	index	found	start	end	mid	list[mid]
21	-1	False	0	15	8	24
			0	7	4	14
			5	7	6	17
21	7	True	7	7	7	21

Implementing a binary search

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Pseudocode

```
FUNCTION binarySearch(list : ARRAY OF INTEGER, item : INTEGER) RETURNS INTEGER
  DECLARE found : BOOLEAN
  DECLARE index : INTEGER
  DECLARE start : INTEGER
  DECLARE end : INTEGER
  DECLARE mid : INTEGER

  found ← FALSE
  index ← -1
  start ← 0
  end ← LENGTH(list) - 1

  WHILE start <= end AND found = FALSE
    mid ← (start + end) DIV 2

    IF list[mid] = item THEN
      found ← TRUE
      index ← mid
    ELSE
      IF list[mid] < item THEN
        start ← mid + 1
      ELSE
        end ← mid - 1
      ENDIF
    ENDIF
  ENDWHILE

  RETURN index
ENDFUNCTION
```

- **Function definition**:
  `FUNCTION binarySearch(list : ARRAY OF INTEGER, item : INTEGER) RETURNS INTEGER`
  - The function searches a **sorted array** for a given item.
  - It returns the **index** of the item if found, or **-1** if not found.
- **Variable declarations**:
  - `found` → a flag to indicate if the item is found (initially `FALSE`)
  - `index` → the result to be returned (initially `-1`)
  - `start` → index of the start of the current search range (initially `0`)
  - `end` → index of the end of the current search range (initially `LENGTH(list) - 1`)
  - `mid` → index of the middle element in the current range
- **Loop condition**:
  `WHILE start <= end AND found = FALSE`
  - Keep searching while the search range is valid **and** the item hasn't been found.

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- **Middle element calculation**:
  mid ← (start + end) DIV 2
  - Calculates the index of the middle element of the current search range.
- **Check for match**:
  IF list[mid] = item THEN
  - If the middle element is equal to the target item:
    - Set found ← TRUE
    - Store the index ← mid
- **Adjust search range**:
  - If the middle value is **less than** the item:
    - Search the **right half** → start ← mid + 1
  - Otherwise:
    - Search the **left half** → end ← mid - 1
- **Loop ends**:
  - The loop stops when the item is found **or** the search range becomes invalid.
- **Return value**:
  RETURN index
  - Returns the index if found, or -1 if not found

Python

```
def binary_search(list, item):
  found = False
  index = -1
  start = 0
  end = len(list) - 1

  while start <= end and not found:
    mid = (start + end) // 2
  if list[mid] == item:
    found = True
    index = mid
  elif list[mid] < item:
    start = mid + 1
  else:
    end = mid - 1

  return index
```

Java

```
public class BinarySearch {

  public static int binarySearch(int[] list, int item) {
     boolean found = false;
     int index = -1;
```

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```
    int start = 0;
    int end = list.length - 1;

    while (start <= end && !found) {  // Corrected found condition
        int mid = (start + end) / 2;

        if (list[mid] == item) {
            found = true;
            index = mid;
        } else if (list[mid] < item) {
            start = mid + 1;
        } else {
            end = mid - 1;
        }
    }

    return index;
}
```

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Bubble sort

What is a bubble sort?

- In A Level Computer Science, a bubble sort sorts items into order, **smallest to largest**
- It **compares pairs of elements** and swaps them if they are out of order
- The first element is compared to the second, the second to the third, the third to the fourth and so on, until the second to last is compared to the last. Swaps occur if each comparison is out of order
- This overall process is called a **pass**

Examiner Tips and Tricks

The highest value in the list eventually "bubbles" its way to the top like a fizzy drink, hence the name "Bubble sort"

- Once the **end of the list** has been reached, the **value at the top of the list is now in order** and the **sort resets back to the start of the list**. The **next largest value** is then **sorted** to the top of the list
- More passes are completed until all elements are in the correct order
- A **final pass** checks all elements and if no swaps are made then the sort is complete
- An example of using a bubble sort would be sorting an array of names into alphabetical order, or sorting an array of student marks from a test

Performing the bubble sort

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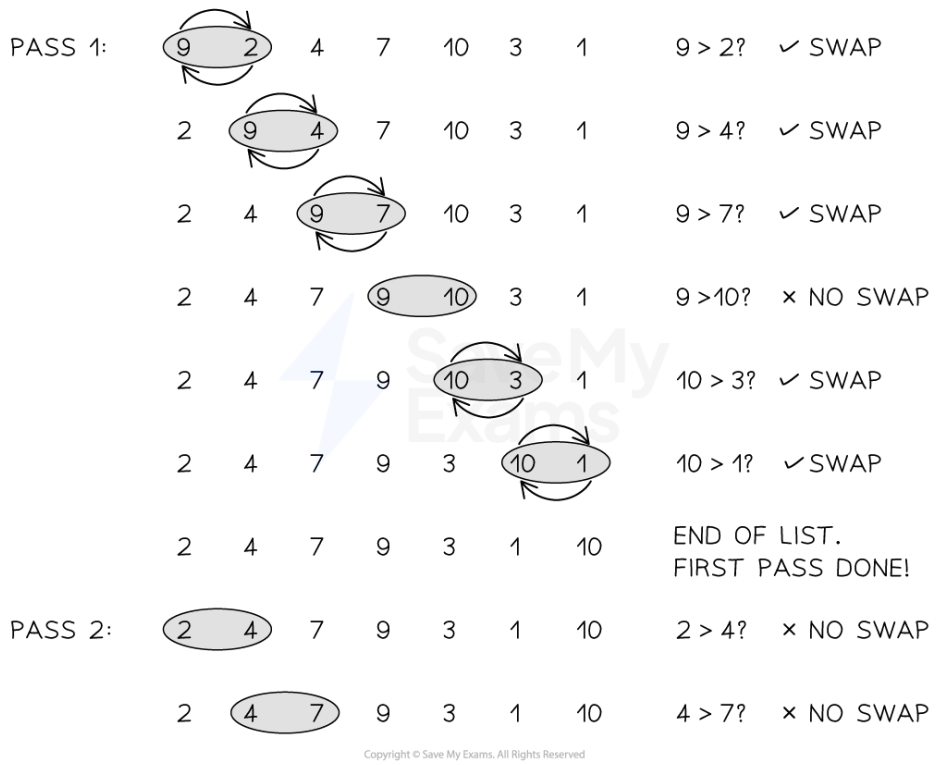

Figure 1: Performing the Bubble sort

Time complexity of the bubble sort

- To determine the algorithm's execution time as measured by the number of instructions or steps carried out Big-O notation must be used
- Four statements are present in the above algorithm O(1), followed by a while loop O(n) containing two statements and a further for loop O(n). This results in the expression: $2n^2 + 4$
- As **coefficients and lower order values are not used**, this give the bubble sort **$O(n^2)$ worst case time complexity**
- The **best case scenario** would be an **already sorted list** which means a time complexity of **O(n)** as each item must still be compared to check if they are in order
- The **average case scenario** would be an **almost sorted list** leading to $O(n^2/2)$, which if coefficients are removed still gives **$O(n^2)$**

Space complexity of the bubble sort

- A bubble sort has a space complexity of O(1) because it operates in-place, requiring a fixed amount of memory
- The fixed amount of memory is for variables such as loop counters and a temporary swap variable

Tracing a bubble sort

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- The trace table follows the algorithm through, updating values that change in the table
- Each iteration of the list reduces the overall size of the list by 1 as after each pass the previously sorted final digit is in order and does not need to be rechecked. This means that 9 is in order and doesn't need to be rechecked, followed by 9 and 7, followed by 9, 7 and 6 and so on
- When the if statement is checked a new row has been added to show the swap of list[j], list[j + 1] and Temp, followed by the subsequent change to Swap from FALSE to TRUE
- After several iterations (that are not shown) the algorithm will eventually output a sorted list

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Trace table of the bubble sort

LastElement	Index	Mylist[Index]	Mylist[Index + 1]	Temp	Swap	Output
10	1	5	9		FALSE	
	2	9	4			
		4	9	4	TRUE	
	3	9	2			
		2	9	9		
	4	9	6			
		6	9			
	5	9	7			
		7	9			
	6	9	1			
		1	9			
	7	9	2			
		2	9			
	8	4	9			

		9	4			
	9	9	3			
		3	9			
9	1	5	4		FALSE	
		4	5	5	TRUE	
	2	5	2			
		2	5			
1					FALSE	1, 2, 2, 3, 4, 4, 5, 6, 7, 9

Your notes

Implementing a bubble sort

Pseudocode

```
DECLARE list : ARRAY[0:9] OF INTEGER
DECLARE last, i, j, temp : INTEGER
DECLARE swap : BOOLEAN

list ← [5, 9, 4, 2, 6, 7, 1, 2, 4, 3]
last ← LENGTH(list)
i ← 0
swap ← TRUE

WHILE i < (last - 1) AND swap = TRUE
   swap ← FALSE

   FOR j ← 0 TO last - i - 2
      IF list[j] > list[j + 1] THEN
         temp ← list[j]
         list[j] ← list[j + 1]
         list[j + 1] ← temp
         swap ← TRUE
      ENDIF
   NEXT j

   i ← i + 1
ENDWHILE

OUTPUT list
```

- **Declare and initialise variables**:

- list is an array of 10 integers to be sorted
- last stores the length of the list
- i is used to track the number of completed passes
- swap is a boolean flag to check if any swaps occurred in a pass

- **Set initial conditions**:
  - i ← 0 → start from the first pass
  - swap ← TRUE → assume a swap will happen to enter the loop
- **Main WHILE loop**:
  - Continues **while** the list still needs sorting (i < last - 1) **and** swaps are still happening (swap = TRUE)
  - At the start of each pass, swap is set to FALSE
- **FOR loop (inner loop)**:
  - Iterates through unsorted part of the list: from index 0 to last - i - 2
  - Compares each pair of adjacent items
- **IF condition**:
  - If the current item is **greater than the next item**, they are swapped
  - swap ← TRUE to indicate a change was made this pass
- **After inner loop**:
  - Increment i to shorten the range on the next pass
- **After sorting**:
  - OUTPUT list displays the sorted list

Python

```
def bubble_sort(list):
    n = len(list)
    for i in range(n - 1):
        swapped = False
        for j in range(0, n - i - 1):
            if list[j] > list[j + 1]:
                list[j], list[j + 1] = list[j + 1], list[j]
                swapped = True
        if not swapped:
            break  # Early termination if no swaps occurred
    return list
```

Java

```
public class BubbleSort {
    public static void bubbleSort(int[] list) {
        int n = list.length;
        for (int i = 0; i < n - 1; i++) {
            boolean swapped = false;
```

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```
    for (int j = 0; j < n - i - 1; j++) {
      if (list[j] > list[j + 1]) {
        int temp = list[j];
        list[j] = list[j + 1];
        list[j + 1] = temp;
        swapped = true;
      }
    }
    if (!swapped) {
      break;  // Early termination if no swaps occurred
    }
  }
}
```

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Worked Example

A global 2D array Result of type INTEGER is used to store a list of exam candidate numbers together with their marks. The array contains 2000 elements, organised as 1000 rows and 2 columns.

Column 1 contains the candidate number and column 2 contains the mark for the corresponding candidate. All elements contain valid exam result data.

A procedure Sort() is needed to sort Result into ascending order of mark using an efficient bubble sort algorithm.

Write pseudocode for the procedure Sort(). **[8]**

Answer

```
PROCEDURE Sort()
  DECLARE Temp : INTEGER
  DECLARE NoSwaps : BOOLEAN
  DECLARE Boundary, Row, Col : INTEGER

  Boundary ← 999
  REPEAT
    NoSwaps ← TRUE
    FOR Row ← 1 TO Boundary
      IF Result[Row, 2] > Result[Row + 1, 2] THEN
        FOR Col ← 1 TO 2
          Temp ← Result [Row, Col]
          Result [Row, Col] ← Result [Row + 1, Col]
          Result [Row + 1, Col] ← Temp
        NEXT Col
        NoSwaps ← FALSE
      ENDIF
    NEXT J
    Boundary ← Boundary - 1
  UNTIL NoSwaps = TRUE
ENDPROCEDURE
```

1. Outer loop [1 mark]
2. Inner loop [1 mark]
3. Correct comparison **in a loop** [1 mark]

4. Correct swap of col1 array elements **in a loop** [1 mark]
5. Correct swap of col2 array elements **in a loop** (via loop or separate statements) [1 mark]
6. 'NoSwap' mechanism: Conditional outer loop including flag reset [1 mark]
7. 'NoSwap' mechanism: Set flag in inner loop to indicate swap [1 mark]
8. Reducing Boundary **in the outer loop** [1 mark]

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Insertion sort

What is an insertion sort?

- In A Level Computer Science, the insertion sort **sorts one item at a time** by **placing it in the correct position** of an unsorted list
- This process repeats **until all items** are in the correct position
- Specifically, the **current item being sorted is compared to each previous item**
- If it is **smaller** than the previous item then the **previous item is moved to the right** and the **current item takes its place**
- If the **current item is larger** than the previous item then it is in the **correct position** and the next current item is then sorted as illustrated below

Performing the insertion sort

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★ 5 IS ALREADY SORTED AS ITS THE FIRST ITEM

5 9 4 2 7 1 2 4 3

9 < 5? × ∴ NO MOVE, NOW SORTED

5 9 4 2 7 1 2 4 3

SORTED LIST SO FAR

---

5 9 4 2 7 1 2 4 3

4 < 9? ✓ ∴ SHIFT 9 TO RIGHT

---

5 4 9 2 7 1 2 4 3

4 < 5? ✓ ∴ SHIFT 5 TO RIGHT

---

4 5 9 2 7 1 2 4 3

SORTED LIST SO FAR

---

4 5 9 2 7 1 2 4 3

2 < 9? ✓ ∴ SHIFT 9 TO RIGHT

2 < 5? ✓ ∴ SHIFT 5 TO RIGHT

2 < 4? ✓ ∴ SHIFT 4 TO RIGHT

2 4 5 9 7 1 2 4 3

SORTED LIST SO FAR

---

2 4 5 9 7 1 2 4 3

7 < 9? ✓ ∴ SHIFT 9 TO RIGHT

7 < 5? × ∴ NO MOVE, NOW SORTED

2 4 5 7 9 1 2 4 3

SORTED LIST SO FAR

ALGORITHM CONTINUES UNTIL ALL SORTED

1 2 2 3 4 4 5 7 9

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Figure 2: Performing the Insertion sort

Time complexity of an insertion sort

- In the above algorithm, one statement is present, a for loop with three statements and a nested while loop that contains two statements
- The time complexity for the insertion sort would be 3n*2n + 1, giving $6n^2 + 1$. 3n comes from the for loop having three statements inside it, not including the while. 2n comes from the while loop having two statements inside it
- Removing coefficients and dominated factors leaves **$O(n^2)$ as the worst case scenario**
- The **best case scenario** would be an **already sorted list** which means each item is checked one at a time giving a time complexity of **O(n)**

The **average case scenario** would be a **half sorted list** giving $O(n^2/2)$, which when coefficients are removed still gives **$O(n^2)$**

Space complexity of an insertion sort

- As the insertion sort requires no additional space, it is space complexity O(1)

Tracing an insertion sort

- Tracing through the insertion sort involves starting at element 1 (not 0)
- i increments through each element one at a time and each item is compared to the one previous. If the previous item is bigger or equal it is set to list[position]
- If position equals 0 then the iteration is at the start of the list and the next iteration begins

Trace Table of the Insertion sort

list	i	item	position	list[position-1]	list[position]
[5, 9, 4, 2, 7, 1, 2, 4, 3]	1	9	1	5	9
	2	4	2	9	9
			1	5	5
			0		
[4, 5, 9, 2, 7, 1, 2, 4, 3]	3	2	3	9	9
			2	5	5
			1	4	4
			0		

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[2, 4, 5, 9, 7, 1, 2, 4, 3]	4	7	4	9	9
			3	5	7
[2, 4, 5, 7, 9, 1, 2, 4, 3]	5	1	5	9	9
			4	7	7
			3	5	5
			2	4	4
			1	2	2
			0		1
[1, 2, 4, 5, 7, 9, 2, 4, 3]	6	2	6	9	9
			5	7	7
			4	5	5
			3	4	4
			2	2	2
[1, 2, 2, 4, 5, 7, 9, 4, 3]	7	4	7	9	9
			6	7	7
			5	5	5
			4	4	4
[1, 2, 2, 4, 4, 5, 7, 9, 3]	8	3	8	9	9
			7	7	7
			6	5	5
			5	4	4

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			4	4	4
			3	2	3
[1, 2, 2, 3, 4, 4, 5, 7, 9]					

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Implementing an insertion sort

Pseudocode

```
PROCEDURE insertionSort(list : ARRAY OF INTEGER)
  DECLARE n, i, position : INTEGER
  DECLARE item : INTEGER

  n ← LENGTH(list)

  FOR i ← 1 TO n - 1
    item ← list[i]
    position ← i

    WHILE position > 0 AND list[position - 1] > item
      list[position] ← list[position - 1]
      position ← position - 1
    ENDWHILE

    list[position] ← item
  NEXT i
ENDPROCEDURE

DECLARE list : ARRAY[0:8] OF INTEGER
list ← [5, 9, 4, 2, 7, 1, 2, 4, 3]

CALL insertionSort(list)

OUTPUT list
```

- **Procedure definition**:
  `insertionSort` takes an array called `list` and sorts it in ascending order
- **Initialisation**:
  - `n ← LENGTH(list)` stores the number of items
  - The sort begins from index `1` because the first item (index 0) is considered sorted
- **Outer FOR loop (`i ← 1 TO n - 1`)**:
  - Each iteration picks the next item in the list to be inserted into the correct position in the sorted portion to its left
- **Store the current item**:
  - `item ← list[i]` stores the current value to be positioned
  - `position ← i` is used to track where `item` should go

- **Inner WHILE loop**:
  - Shifts elements in the sorted portion **to the right** while they are greater than item
  - Stops when either:
    - The start of the list is reached (position = 0)
    - Or the correct spot is found (list[position - 1] <= item)
- **Insert the item**:
  - list[position] ← item inserts the value in the correct position
- **Repeat**:
  - The next item is considered, and the process repeats until the entire list is sorted
- **After sorting**:
  - OUTPUT list displays the sorted array

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Python

```
def insertion_sort(data):
    for i in range(1, len(data)):
        item = data[i]
        position = i
        while position > 0 and data[position - 1] > item:
            data[position] = data[position - 1]
            position -= 1
        data[position] = item
    return data
```

Java

```
public class InsertionSort {

    public static void insertionSort(int[] data) {
        for (int i = 1; i < data.length; i++) {
            int item = data[i];
            int position = i;
            while (position > 0 && data[position - 1] > item) {
                data[position] = data[position - 1];
                position--;
            }
            data[position] = item;
        }
    }
```

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Implementing a stack

How do you program a stack?

- To recap the main operations of stacks and how they work, read the ADT overview
- When programming a stack, the main operations listed below must be included in your program.
- The main operations for stacks are:

Main operations

Operation	Pseudocode	Python	Java
isEmpty()	FUNCTION isEmpty(s : Stack) RETURNS BOOLEAN RETURN s.top = -1 ENDFUNCTION	def isEmpty(self): return len(self.stack) == 0	stack.isEmpty();
isFull()	FUNCTION isFull(s : Stack) RETURNS BOOLEAN RETURN s.top = s.capacity - 1 ENDFUNCTION	def isFull(self): return len(self.stack) == self.capacity	public boolean isFull() { return top == capacity - 1; }
push(value)	PROCEDURE push(s : REF Stack, value : INTEGER) IF NOT isFull(s) THEN s.top ← s.top + 1 s.items[s.top] ← value ELSE OUTPUT "Stack is full" ENDIF ENDPROCEDURE	stack.push(1)	stack.push(1);
pop()	FUNCTION pop(s : REF Stack) RETURNS INTEGER IF NOT isEmpty(s) THEN DECLARE value : INTEGER value ← s.items[s.top] s.top ← s.top - 1 RETURN value ELSE RETURN -1 ENDIF ENDFUNCTION	popped_item = stack.pop()	int poppedItem = stack.pop();
peek()	FUNCTION peek(s : Stack) RETURNS INTEGER IF NOT isEmpty(s) THEN RETURN s.items[s.top] ELSE RETURN -1 ENDIF ENDFUNCTION	def peek(self): if not self.isEmpty(): return self.stack[-1] else: return None	int topItem = stack.peek();
size()	FUNCTION size(s : Stack) RETURNS INTEGER RETURN s.top + 1 ENDFUNCTION	size = len(stack)	int size = stack.size();

- In **pseudocode**, we assume a record-based stack with fields like items, top, and capacity

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- Python uses built-in lists or a class with .push() and .pop() methods
- Java uses Stack<E> from java.util.Stack

Your notes

Stack in operation (Pseudocode)

```
TYPE Stack
   DECLARE items : ARRAY[0:4] OF INTEGER
   DECLARE top : INTEGER
   DECLARE capacity : INTEGER
ENDTYPE

// Initialise stack
DECLARE s : Stack
s.top ← -1
s.capacity ← 5

// Push 3 values onto the stack
CALL push(s, 10)
CALL push(s, 20)
CALL push(s, 30)

// Peek at the top item
OUTPUT "Top item is ", peek(s)

// Pop 2 values from the stack
DECLARE val : INTEGER
val ← pop(s)
OUTPUT "Popped value: ", val

val ← pop(s)
OUTPUT "Popped value: ", val

// Output current stack size
OUTPUT "Current size: ", size(s)
```

- **Defines a Stack structure** with a fixed-size array of 5 items, a top pointer, and a capacity
- **Initialises the stack** with top ← -1 to indicate it is empty
- **Pushes 3 values** (10, 20, 30) onto the stack one by one:
  - Each push increases top and adds the value to the items array
- **Peeks** at the current top of the stack:
  - Returns 30 without removing it
- **Pops two values** off the stack:
  - First pop returns 30, second returns 20
  - Each pop decreases top by 1
- **Outputs the current size** of the stack:
  - After popping twice, only one value (10) remains → size is 1

Python

```
class Stack:
    def __init__(self, capacity):
        self.stack = []
        self.capacity = capacity

    def isEmpty(self):
        return len(self.stack) == 0

    def isFull(self):
        return len(self.stack) == self.capacity

    def push(self, value):
        if not self.isFull():
            self.stack.append(value)
        else:
            print("Stack is full")

    def pop(self):
        if not self.isEmpty():
            return self.stack.pop()
        else:
            return None

    def peek(self):
        if not self.isEmpty():
            return self.stack[-1]
        else:
            return None

    def size(self):
        return len(self.stack)


# Stack in operation
s = Stack(5)

# Push 3 values
s.push(10)
s.push(20)
s.push(30)

# Peek at top
print("Top item is", s.peek())

# Pop 2 values
print("Popped value:", s.pop())
print("Popped value:", s.pop())

# Output current size
print("Current size:", s.size())
```

Java

```
import java.util.Stack;

public class StackExample {
    public static void main(String[] args) {
        Stack<Integer> s = new Stack<>();
```

Your notes

```
    // Push 3 values
    s.push(10);
    s.push(20);
    s.push(30);

    // Peek at top
    System.out.println("Top item is " + s.peek());

    // Pop 2 values
    System.out.println("Popped value: " + s.pop());
    System.out.println("Popped value: " + s.pop());

    // Output current size
    System.out.println("Current size: " + s.size());
  }
}
```

Your notes

Worked Example

Study the pseudocode function PushAnimal():

```
FUNCTION PushAnimal(DataToPush : STRING) RETURNS BOOLEAN
  IF AnimalTopPointer = 20 THEN
    RETURN FALSE
  ELSE
    Animal[AnimalTopPointer] ← DataToPush
    AnimalTopPointer ← AnimalTopPointer + 1
    RETURN TRUE
  ENDIF
ENDFUNCTION
```

Write program code for the function PushAnimal() **[3]**

Answer

- Function header (and close where appropriate) with parameter, checking if full (AnimalTopPointer = 20) **and** returning false [1 mark]
- If not full, inserting parameter value into AnimalTopPointer [1 mark]
- ...incrementing pointer and returning true [1 mark]

Example program code:

Java

```
public static Boolean PushAnimal(String DataToPush){
  if(AnimalTopPointer == 20){
    return false;
  }else{
    Animal[AnimalTopPointer] = DataToPush;
    AnimalTopPointer++;
    return true;
  }
}
```

VB.NET

```
Function PushAnimal(DataToPush)
  If AnimalTopPointer = 20 Then
    Return False
  Else
    Animal(AnimalTopPointer) = DataToPush
    AnimalTopPointer = AnimalTopPointer + 1
    Return True
  End If
End Function
```

Python

```
def PushAnimal(DataToPush):
  global AnimalTopPointer
  global ColourTopPointer
  if AnimalTopPointer == 20:
    return False
  else:
    Animal.append(DataToPush)
    AnimalTopPointer +=1
    return True
```

Your notes

Your notes

Implementing a queue

How do you program a queue?

- A **queue** is a **linear ADT** that follows the **First In, First Out (FIFO)** principle:
  - **Enqueue**: Add item to the back
  - **Dequeue**: Remove item from the front
  - **Peek/front**: Look at the front item without removing
  - **isEmpty / isFull**: Check queue status
- Assuming a **record-based circular queue** implementation:

Enqueue

```
PROCEDURE enqueue(q : REF Queue, item : STRING)
   IF NOT isFull(q) THEN
      q.rear ← (q.rear + 1) MOD q.capacity
      q.items[q.rear] ← item
      q.size ← q.size + 1
   ELSE
      OUTPUT "Queue is full"
   ENDIF
ENDPROCEDURE
```

Dequeue

```
FUNCTION dequeue(q : REF Queue) RETURNS STRING
   IF NOT isEmpty(q) THEN
      DECLARE value : STRING
      value ← q.items[q.front]
      q.front ← (q.front + 1) MOD q.capacity
      q.size ← q.size - 1
      RETURN value
   ELSE
      RETURN "Queue is empty"
   ENDIF
ENDFUNCTION
```

isEmpty / isFull

```
FUNCTION isEmpty(q : Queue) RETURNS BOOLEAN
   RETURN q.size = 0
ENDFUNCTION

FUNCTION isFull(q : Queue) RETURNS BOOLEAN
   RETURN q.size = q.capacity
ENDFUNCTION
```

Find (linear search)

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```
FUNCTION find(q : Queue, target : STRING) RETURNS BOOLEAN
   DECLARE i : INTEGER
   DECLARE pos : INTEGER
   pos ← q.front

   FOR i ← 1 TO q.size
      IF q.items[pos] = target THEN
         RETURN TRUE
      ENDIF
      pos ← (pos + 1) MOD q.capacity
   NEXT i

   RETURN FALSE
ENDFUNCTION
```

Python

```
from collections import deque

class Queue:
   def __init__(self, capacity):
      self.items = deque()
      self.capacity = capacity

   def isEmpty(self):
      return len(self.items) == 0

   def isFull(self):
      return len(self.items) == self.capacity

   def enqueue(self, item):
      if not self.isFull():
         self.items.append(item)
      else:
         print("Queue is full")

   def dequeue(self):
      if not self.isEmpty():
         return self.items.popleft()
      else:
         return "Queue is empty"

   def peek(self):
      if not self.isEmpty():
         return self.items[0]
      else:
         return None

   def find(self, target):
      return target in self.items

# Example usage
q = Queue(5)
q.enqueue("Alice")
q.enqueue("Bob")
q.enqueue("Charlie")
print("Front item:", q.peek())
print("Popped:", q.dequeue())
print("Is 'Bob' in queue?", q.find("Bob"))
```

Your notes

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Java

```
import java.util.LinkedList;
import java.util.Queue;

public class QueueExample {
   public static void main(String[] args) {
      Queue<String> q = new LinkedList<>();

      // Enqueue
      q.offer("Alice");
      q.offer("Bob");
      q.offer("Charlie");

      // Peek
      System.out.println("Front item: " + q.peek());

      // Dequeue
      System.out.println("Popped: " + q.poll());

      // Find
      System.out.println("Is 'Bob' in queue? " + q.contains("Bob"));
   }
}
```

Your notes

Your notes

Linked lists

How do you program a linked list?

- To recap the main operations of linked lists and how they work, read our ADT Overview page
- To program a Linked List you must be able to perform the following operations:
  - **Create a linked list**
  - **Traverse a linked list**
  - **Add data to a linked list**
  - **Remove data from a linked list**

Create a linked list

- Define the class called 'Node' that represents a node in the linked list.
- Each node has 2 fields 'fruit' (this stores the fruit value) and 'next' (this stores a reference to the next node in the list)
- We create 3 nodes and assign fruit values to each node
- To establish a connection between the nodes we set the 'next' field of each node to the appropriate next node

Pseudocode

```
CLASS Node
   DECLARE fruit : STRING
   DECLARE next : Node
ENDCLASS

// Create nodes
DECLARE node1 : Node
DECLARE node2 : Node
DECLARE node3 : Node

SET node1 = NEW Node
SET node2 = NEW Node
SET node3 = NEW Node

// Assign fruit values
SET node1.fruit = "apple"
SET node2.fruit = "banana"
SET node3.fruit = "orange"

// Connect nodes
SET node1.next = node2
```


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```
SET node2.next = node3
SET node3.next = NULL
```

Your notes

Python

```python
class Node:
 def __init__(self, fruit):
   self.fruit = fruit
   self.next = None

# Create nodes
node1 = Node("apple")
node2 = Node("banana")
node3 = Node("orange")

# Connect nodes
node1.next = node2
node2.next = node3
```

Java

```java
class Node {
   String fruit;
   Node next;

   Node(String fruit) {
      this.fruit = fruit;
      this.next = null;
   }
}

public class LinkedListExample {
   public static void main(String[] args) {
      // Create nodes
      Node node1 = new Node("apple");
      Node node2 = new Node("banana");
      Node node3 = new Node("orange");

      // Connect nodes
      node1.next = node2;
      node2.next = node3
   }
}
```

Algorithm to traverse a linked list

- We traverse the linked list starting from 'node1'. We output the fruit value of each node and update the 'current' reference to the next node until we reach the end of the list

Pseudocode

```
// Start at the head of the list
DECLARE current : Node
SET current = node1

WHILE current != NULL
   OUTPUT current.fruit
   SET current = current.next
ENDWHILE
```

Python

```
# Traverse the linked list

current = node1

while current is not None:

   print(current.fruit)

   current = current.next
```

Java

```
// Traverse the linked list

current = node1;

while (current != null) {

System.out.println(current.fruit);

    current = current.next;
```

Algorithm to add data to a linked list

- We create a new node with the new data and check if the list is empty
- If it isn't, we find the last node and add the new node to the end

Pseudocode

```
// Create a new node
DECLARE newNode : Node
SET newNode = NEW Node
SET newNode.fruit = "orange"
SET newNode.next = NULL

// If the list is empty
IF node1 = NULL THEN
   SET node1 = newNode
ELSE
   // Traverse to the end of the list
```

Your notes

```
    DECLARE current : Node
    SET current = node1

    WHILE current.next != NULL
        SET current = current.next
    ENDWHILE

    // Append the new node
    SET current.next = newNode
ENDIF
```

Python

```
# Add "orange" to the end of the linked list
new_node = Node("orange")

if node1 is None:
    # If the list was empty, make "orange" the head
    node1 = new_node
else:
    # Find the last node and append "orange" to it
    current = node1
    while current.next is not None:
        current = current.next
    current.next = new_node
```

Java

```
// Add "orange" to the end of the linked list
Node new_node = new Node("orange");

if (node1 == null) {
// If the list was empty, make "orange" the head
node1 = new_node;
} else {
// Find the last node and append "orange" to it
    current = node1;
    while (current.next != null) {
    current = current.next;
    }
    current.next = new_node;

}
```

Algorithm to remove data from a linked list

- We traverse the list until we find the data to be removed
- If the data to be removed is the first node, we update the head, else we update the node before to bypass the data to be removed

Your notes

Pseudocode

```
// Start traversal from the head
DECLARE current : Node
DECLARE previous : Node

SET current = node1
SET previous = NULL

WHILE current != NULL
  IF current.fruit = "apple" THEN
    IF previous = NULL THEN
      // "apple" is at the head of the list
      SET node1 = current.next
    ELSE
      // Bypass the current node
      SET previous.next = current.next
    ENDIF
    BREAK
  ENDIF

  SET previous = current
  SET current = current.next
ENDWHILE
```

Python

```
# Find and remove "apple"

while current is not None:

  if current.fruit == "apple":

    if previous is None:

      # If "apple" is the first node, update the head

      node1 = current.next

    else:

      # Remove the node by updating the previous node's next pointer

      previous.next = current.next

    break

  previous = current

  current = current.next
```

Java

```
// Find and remove "apple"
```

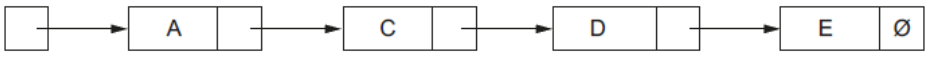
Your notes

```
while (current != null) {
 if (current.fruit.equals("apple")) {
if (previous == null) {
            // If "apple" is the first node, update the head
            node1 = current.next;
           } else {
            // Remove the node by updating the previous node's next pointer
            previous.next = current.next;
            }
            break;
         }
previous = current;
     current = current.next;
   }
```

Your notes

Worked Example

The following diagram represents an Abstract Data Type (ADT) for a linked list.



The free list is as follows:

Explain how a node containing data value B is added to the list in alphabetic sequence.**[4]**

Answer

One mark per point

1. Check for a free node [1 mark]
2. Search for correct insertion point [1 mark]

3. Assign data value B to first node in free list / node pointed to by start pointer of free list [1 mark]
4. Pointer from A will be changed to point to node containing B (instead of C) [1 mark]
5. Pointer from B will be changed to point to node containing C [1 mark]
6. Start pointer in free list moved to point to next free node [1 mark]

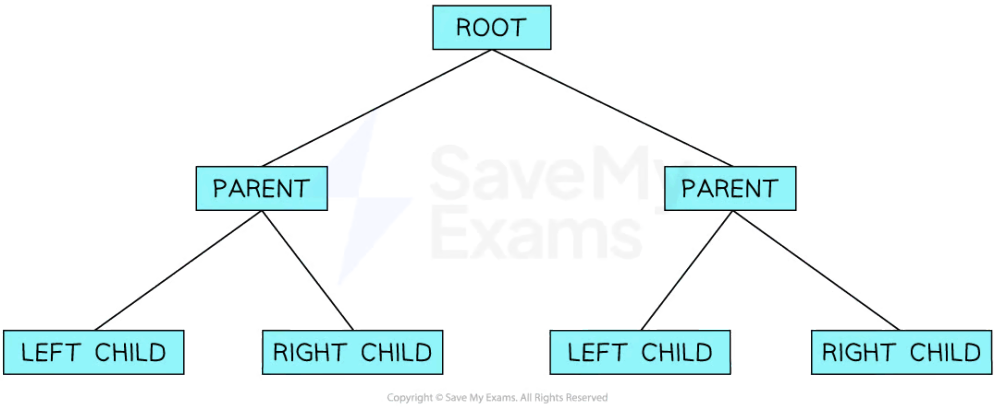
Your notes

Your notes

Binary trees

What is a binary tree?

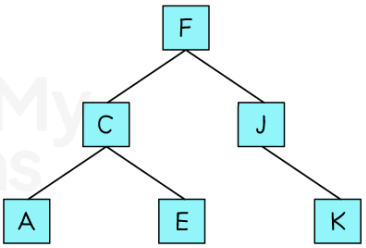
- A binary tree is a **rooted tree** where every node has a **maximum of 2 nodes**
- A binary tree is essentially a **graph** and therefore can be implemented in the same way
- You must understand:
  - **Tree traversal of a tree data structure**
  - **Add new data to a tree**
  - **Remove data from a tree**



- The most common way to represent a binary tree is by storing each node with a **left and right pointer**

This information is usually implemented **using 2D arrays**

Index	Left Pointer	Data Value	Right Pointer
0	1	F	3
1	2	C	4
2	–	A	–
3	–	J	5
4	–	E	–
5	–	K	–



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Your notes

Tree terminology

Keyword	Definition
Node	An item in a tree
Edge	Connects two nodes together and is also known as a branch or pointer
Root	A single node which does not have any incoming nodes
Child	A node with incoming edges
Parent	A node with outgoing edges
Subtree	A subsection of a tree consisting of a parent and all the children of a parent
Leaf	A node with no children
Traversing	The process of visiting each node in a tree data structure, exactly once

How do you program a tree?

- In the following example:
  - A tree is represented using a TreeNode **structure**, which includes:
    - A value field to store the data held in the node
    - A children field to store a collection (e.g. list or array) of child nodes

- The createTreeNode function is used to **initialise a new tree node**, setting its value and creating an empty collection for its children
- The addChild function allows a new child node to be **added to a parent node**
  - It creates a new TreeNode with the given value and appends it to the parent's children collection
- Finally, the createTreeNode function is used to create the **root node**
  - Additional child nodes are added using the addChild function, allowing a **multi-level tree structure** to be built with branches and sub-branches

Your notes

Pseudocode

```
CLASS TreeNode
   DECLARE value : STRING
   DECLARE children : ARRAY OF TreeNode
ENDCLASS

FUNCTION createTreeNode(val : STRING) RETURNS TreeNode
   DECLARE node : TreeNode
   SET node = NEW TreeNode
   SET node.value = val
   SET node.children = []
   RETURN node
ENDFUNCTION

PROCEDURE addChild(parent : TreeNode, childValue : STRING)
   DECLARE childNode : TreeNode
   SET childNode = createTreeNode(childValue)
   APPEND childNode TO parent.children
ENDPROCEDURE

// Example usage
DECLARE root : TreeNode
SET root = createTreeNode("Root")

CALL addChild(root, "Child 1")
CALL addChild(root, "Child 2")

// Adding a grandchild to Child 1
DECLARE child1 : TreeNode
SET child1 = root.children[0]
CALL addChild(child1, "Grandchild 1.1")
```

Python

```
class TreeNode:
   def __init__(self, value):
      self.value = value
      self.children = []

def createTreeNode(val):
   return TreeNode(val)
```

```
def addChild(parent, childValue):
    childNode = createTreeNode(childValue)
    parent.children.append(childNode)

# Example usage
root = createTreeNode("Root")

addChild(root, "Child 1")
addChild(root, "Child 2")

# Adding a grandchild to "Child 1"
child1 = root.children[0]
addChild(child1, "Grandchild 1.1")
```

Java

```
import java.util.ArrayList;

class TreeNode {
    String value;
    ArrayList<TreeNode> children;

    // Constructor
    public TreeNode(String value) {
        this.value = value;
        this.children = new ArrayList<>();
    }
}

public class TreeExample {

    // Function to create a tree node
    public static TreeNode createTreeNode(String val) {
        return new TreeNode(val);
    }

    // Function to add a child to a parent node
    public static void addChild(TreeNode parent, String childValue) {
        TreeNode childNode = createTreeNode(childValue);
        parent.children.add(childNode);
    }

    public static void main(String[] args) {
        // Create the root node
        TreeNode root = createTreeNode("Root");

        // Add children to the root
        addChild(root, "Child 1");
        addChild(root, "Child 2");

        // Add a grandchild to "Child 1"
        TreeNode child1 = root.children.get(0);
        addChild(child1, "Grandchild 1.1");
    }
}
```

Your notes

Algorithm to traverse a tree structure

Your notes

Pseudocode – pre-order tree traversal

```
PROCEDURE traverseTree(node : TreeNode)
  IF node = NULL THEN
    RETURN
  ENDIF

  OUTPUT node.value

  FOR EACH child IN node.children DO
    CALL traverseTree(child)
  NEXT
ENDPROCEDURE

// Call the traversal on the root
CALL traverseTree(root)
```

Python – pre-order tree traversal

```
def traverseTree(node):
  if node is None:
    return

  print(node.value)

  for child in node.children:
    traverseTree(child)

# Call the traversal
traverseTree(root)
```

Java – pre-order tree traversal

```
public static void traverseTree(TreeNode node) {
  if (node == null) {
    return;
  }

  System.out.println(node.value);

  for (TreeNode child : node.children) {
    traverseTree(child);
  }
}

// In main:
traverseTree(root);
```

- For a tree like:

```
Root
├── Child 1
│   └── Grandchild 1.1
└── Child 2
```

- The output will be:

```
Root
Child 1
Grandchild 1.1
Child 2
```

Algorithm to add data to a tree structure

Pseudocode

```
PROCEDURE addChild(parent : TreeNode, childValue : STRING)
   DECLARE childNode : TreeNode
   SET childNode = createTreeNode(childValue)
   APPEND childNode TO parent.children
ENDPROCEDURE

// Usage:
CALL addChild(root, "New Child")
```

Python

```
def addChild(parent, childValue):
   childNode = TreeNode(childValue)
   parent.children.append(childNode)

# Usage:
addChild(root, "New Child")
```

Java

```
public static void addChild(TreeNode parent, String childValue) {
   TreeNode childNode = new TreeNode(childValue);
   parent.children.add(childNode);
}

// Usage in main:
addChild(root, "New Child");
```

- These examples all **create a new node** with the provided value and **attach it to the given parent node**
- This approach allows dynamic construction of trees with multiple levels

Your notes

Algorithm to remove data to a tree structure

Your notes

Pseudocode

```
PROCEDURE removeChild(parent : TreeNode, targetValue : STRING)
  FOR index FROM 0 TO LENGTH(parent.children) - 1
    IF parent.children[index].value = targetValue THEN
      REMOVE parent.children[index]
      RETURN
    ENDIF
  NEXT
ENDPROCEDURE

// Usage
CALL removeChild(root, "Child 1")
```

Python

```
def removeChild(parent, targetValue):
  for i in range(len(parent.children)):
    if parent.children[i].value == targetValue:
      del parent.children[i]
      return  # Exit after removing first match

# Usage:
removeChild(root, "Child 1")
```

Java

```
public static void removeChild(TreeNode parent, String targetValue) {
  for (int i = 0; i < parent.children.size(); i++) {
    if (parent.children.get(i).value.equals(targetValue)) {
      parent.children.remove(i);
      return; // Exit after first match
    }
  }
}

// Usage in main:
removeChild(root, "Child 1");
```

- This only removes the **first matching child** from the direct children of a node
- If you want to remove nodes **anywhere in the tree**, you'd need a recursive version
- REMOVE and del and .remove() work because the children collection is a list/array

Your notes

ADT from ADT

What does it mean?

- You should be able to **explain or demonstrate** how you can build one abstract data type **on top of another**
- For example:
  - A **stack** can be implemented using a **linked list**
  - A **queue** can be implemented using two **stacks**
  - A **dictionary** can be implemented using a **binary search tree**
- You're not just limited to built-in types like arrays, you can **compose ADTs out of each other**

What is an ADT?

- If you are unsure what an ADT is, read our ADT overview page

How are ADTs implemented?

Using built-in types (e.g. arrays, lists)

ADT	Built-in type used	Example
Stack	List/array	Use .append() and .pop() in Python
Queue	List/array or circular buffer	Use .append() and .pop(0)
Linked list	Class with next fields	Implement nodes manually
Dictionary	Hash table (built-in dict)	Key-value mapping
Binary tree	Class with left and right	Each node links to children

Using other ADTs

ADT being built	Uses this ADT	Example
Queue	Two stacks	Enqueue in one stack, dequeue using a second stack

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Stack	Linked list	Push/pop from head of list
Binary tree	Linked list or array	Nodes link to each other like a graph
Dictionary	Binary search tree	Store keys and values in BST for ordered lookup

Your notes

Example: Implementing a Queue using Two Stacks

Idea:

- Use **two stacks** (let's say inStack and outStack) to simulate a queue

How it works:

- enqueue(x) → push x onto inStack
- dequeue() → if outStack is empty, pop all items from inStack and push onto outStack, then pop from outStack
- This simulates **FIFO behaviour** using **LIFO operations** from stacks

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Suitability of algorithms

Why do we need to check the suitability of algorithms?

- Algorithms must be compared to **determine how suitable they are for solving a specific problem** or working with a particular set of data
- Key factors in comparing algorithms include:
  - **How long** they take to complete a task (**time complexity**)
  - **How much memory** they require to complete a task (**space complexity**)
- Generally, the most suitable algorithm is the one that solves the problem in the **shortest time** and using the **least memory**

Time complexity

- Time complexity refers to the **number of operations or steps** an algorithm takes to complete, **not the actual time in seconds or minutes**
- It is important to understand that time complexity is **independent of hardware performance**
- A faster CPU may execute instructions more quickly, but the number of instructions the algorithm performs remains the same

Analogy: Running a marathon

- The marathon distance = the algorithm
- The runners = different CPUs
- Each runner finishes at a different time, but the distance (number of steps) is the same

Space complexity

- Space complexity refers to the **amount of memory** an algorithm needs to complete its task
- Memory usage includes:
  - Space for input data
  - Temporary variables
  - Any additional data structures (e.g. arrays, stacks, queues)
- For example:
  - If an algorithm **creates a new copy** of the input data, it will require more memory than one that modifies the data in place
  - An algorithm that uses **recursion** may use more memory due to the **call stack**

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Big O Notation

What is Big O Notation?

- Big O Notation is a mathematical way to describe the **time** and **space complexity** of an algorithm
- It shows how well an algorithm scales as the size of the input increases
  - It describes the **order of growth** (efficiency) of an algorithm
  - It is **hardware-independent** – it measures steps or operations, not seconds
  - Named after German mathematician **Edmund Landau**

Big O Rules:

1. Keep only the **dominant term** (largest exponent or growth rate)
2. **Ignore constants** – they become insignificant as input size grows

Types of time complexity

O(1) – Constant time

- The algorithm always takes the same number of steps, regardless of input size

```
totalLength = len(s)
```

- The length is returned in one step, even if the string is very long

O(n) – Linear time

- The number of steps increases proportionally with the input size n

```
function sumNumbers1(n)
  sum = 0
  for i = 1 to n
    sum = sum + i
  next i
  return sum
endfunction
```

- For input size n, the algorithm performs n + 1 steps
- Example with multiple loops:

```
function loop
  x = input
  for i = 1 to x
    print(x)
  next i

  a = 0
  for j = 1 to x
    a = a + 1
  next j
  print(a)
```

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```
  b = 1
  for k = 1 to x
    b = b + 1
  next k
  print(b)
endfunction
```

- Total steps = 3x + 5 ⇒ Still O(n) as x grows

O(n^2) – Polynomial time

- Performance is proportional to the square of the input size
- Caused by **nested loops**

```
function timesTable
  x = 12
  for i = 1 to x
    for j = 1 to x
      print(i, " x ", j, " = ", i*j)
    next j
  next i

  for k = 1 to x
    print(k)
  next k

  print("Hello World!")
endfunction
```

- Steps: x^2 + x + 2 ⇒ Dominant term is x^2 ⇒ O(n^2)
- As x increases, the impact of lower-order terms and constants becomes negligible

O(log n) – Logarithmic time

- The number of steps grows slowly even as input size grows rapidly
- Common in **binary search**
  - Example: log2(8) = 3, because 2^3 = 8
  - Doubling input size barely affects total steps

x	log2(x)
1	0
8	3
1,024	10
1,048,576	20

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1,073,741,824	30

- Very efficient – useful for large datasets

O(2^n) – Exponential time

- The number of steps **doubles** for every additional input value
- Grows extremely fast
- Very inefficient, but some problems (e.g. brute-force solutions) fall into this category

x	2^x
1	2
10	1,024
100	~10^30

- Rare in practice unless necessary

O(n log n) – Linear logarithmic time

- Algorithms that **divide** data and **process** each part
- Example: **Merge Sort**
- More efficient than O(n^2), commonly used in sorting

Best, average, and worst case

- Every algorithm has different performance depending on input:

Case	Description	Example (Linear Search)
Best case	Most efficient scenario	Item is first → O(1)
Average case	Typical input	Item is in middle → O(n)
Worst case	Least efficient scenario	Item not found → O(n)

- **Big O notation usually describes the worst-case** performance to guarantee a minimum standard

How to derive Big O from code

1. **Count loops**:
    - 1 loop = O(n)

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- 2 nested loops = O(n^2)
- No loops = O(1)

2. **Ignore constants and lower-order terms**:
   - 3n^2 + 4n + 5 → O(n^2)

Examples

```
function sumNumbers1(n)
  sum = 0
  for i = 1 to n
    sum = sum + i
  next i
  return sum
endfunction
```

- One loop → O(n)

```
function sumNumbers2(n)
  sum = n * (n + 1) / 2
  return sum
endfunction
```

- One statement → O(1)

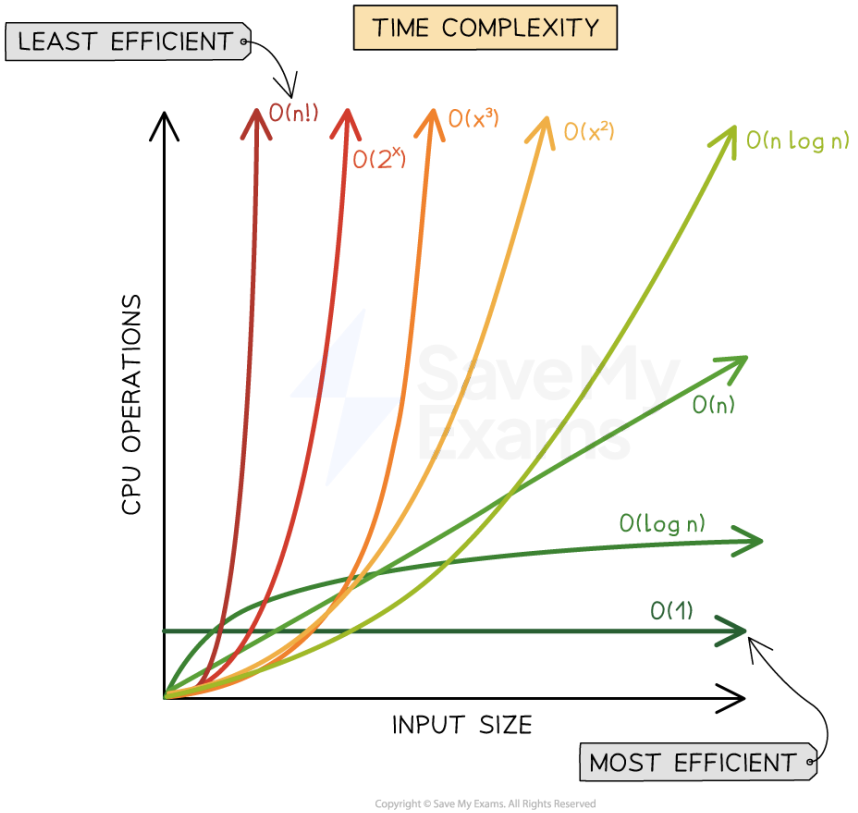
```
function loop
  (3 loops, 5 statements)
  ⇒ O(n)
endfunction

function timesTable
  (nested loop + 1 loop + 1 statement)
  ⇒ O(n^2)
endfunction
```

Time complexity graph

Your notes

Your notes



- From the graph, **exponential time (2^n)** grows the **quickest** and therefore is the most **inefficient** form of algorithm, followed by **polynomial time (n^2)**, **linear time (n)**, **logarithmic time (logn)** and finally **constant time (1)**
- Further time complexities exist such as **factorial time (n!)** which performs **worse than exponential time** and **linear-logarithmic time (nlogn)** which sits **between polynomial and linear time**. Merge sort is an example of an algorithm that uses O(nlogn) time complexity
- The Big-O time complexity of an algorithm can be derived by inspecting its component parts
- It is important to know that when determining an algorithm's Big-O that **only the dominant factor is used**.
- This means the time complexity that **contributes the most** to the algorithm is used to describe the algorithm. For example:
  - $3x^2 + 4x + 5$, which describes three nested loops, four standard loops and five statements is described as $O(n^2)$
  - x^2 is the dominating factor. As the input size grows x^2 contributes proportionally more than the other factors
  - Coefficients are also removed from the Big-O notation. The 3 in $3x^2$ acts as a multiplicative value whose contribution becomes smaller and smaller as the size of the input increases.

- For an arbitrarily large value such as 10^{100}, which is 10 with one hundred 0's after it, multiplying this number by 3 will have little contribution to the overall time complexity. 3 is therefore dropped from the overall time complexity

Your notes